In-Semester Exam-I (Autumn'2026)

DS604 Introduction to Data Management

Time: 60 Minutes | Max Points: 70

IMPORTANT NOTE:
1. Write answers neat and clean. Answers that are difficult to read will simply be discarded.
2. All questions carry equal marks
3. In all questions marks awarding strategy will be discrete (i.e., 0, half, and full marks).
4. You may have to pay penalty for lengthier solutions.

Consider the schema given below. It is almost the same as the one used in one of the labs. However, there may be some changes here.

```
COURSE(CNO, CName, Credits)
    PK: CNO
INSTRUCTOR(IID, IName)
    PK: IID
PROGRAM(PID, PName)
    PK: PID
BATCH(BID, BYEAR, PID)
    // Each tuple represent a batch of a program, and attribute
    // BYEAR specifies Admission Year of the batch
    // Every admission year adds one entry
    PK: BID
    FK: PID REFERENCES PROGRAM(PID)
STUDENT(SID, Name, BID, CPI)
    PK: SID
    FK: BID REFERENCES STUDENT(BID)
SEMESTER(SemID, AcadYear, Sem)
    PK: SemID
OFFERS(OfferID, SemID, CNO, CType, IID)
    // Attributes CType specifies, if course offered is
    // CORE or ELECTIVE
    PK: Offer-ID
    FK: SemID REFERENCES SEMESTER(SemID)
    FK: CNO REFERENCES COURSE(CNO)
    FK: IID REFERENCES INSTRUCTOR(IID)
OFFERED_FOR(OfferID, BID)
    // specifies what course is available for students of
    // what batch of students, and attributes BID specifies
    // batch for which the course is offered
    PK: (OfferID, BID)
    FK: BID REFERENCES BATCH(BID)
REGISTERS(SID, OfferID, Grade)
    PK: {SID, OfferID}
    FK: OfferID REFERENCES OFFERS(OfferID)
RESULT(SemID, SID, SPI)
    // stores SPI of students of every semester
    PK: (SemID, SID)
```

*Write **Relational Algebra** expressions to answer the following queries.*

1. List students (Cols: SID, SName) from students of batch=2025 of Prog-Id='12' that are having (CPI < 5 OR SPI in Autumn'2025-26 < 6) .

$$r1 \leftarrow RESULT * SEMESTER * STUDENT * BATCH$$
$$r2 \leftarrow \sigma_{PID='12' \text{ AND } ACADYR=2025 \text{ AND } SEM='AUTUMN' \text{ AND } BYEAR=2025}(r1)$$
$$result \leftarrow \pi_{SID,SNAMEE}(r2)$$

2. List ELECTIVE (CTYPE='ELECTIVE') courses that have been offered in WINTER '2025-26 with their credits, and Instructor Name; that is, we have columns (CNO, CName, Credits, Instructor-Name, PID, and Batch) in the result.

$$r1 \leftarrow OFFERED_FOR * OFFERS * SEMESTER * BATCH * INSTRUCTOR$$
$$r2 \leftarrow \sigma_{CTYPE='ELECTIVE' \text{ AND } ACADYEAR=2025 \text{ AND } SEM='WINTER'}(r1)$$
$$result \leftarrow \pi_{CNO,CNAME,CREDITS,INAME,PID,BYEAR}(r2)$$

3. List CORE (CTYPE='CORE') courses (CNO, CName, Credits, Instructor-Name) that are offered for Prog-ID = '18' in Autumn'2025-26.

$$r1 \leftarrow OFFERED_FOR * OFFERS * SEMESTER * BATCH * INSTRUCTOR$$
$$r2 \leftarrow \sigma_{CTYPE='CORE' \text{ AND } PID='18' \text{ AND } ACADYEAR=2025 \text{ AND } SEM='AUTUMN'}(r1)$$
$$result \leftarrow \pi_{CNO,CNAME,CREDITS,INAME}(r2)$$

4. List students (cols: SID, SName) of Prog-ID=11 and batch 2025 who have registered in course CNO='IT494' in AUTUMN '2026-27.

$$r1 \leftarrow REGISTERS * OFFERS * SEMESTER * BATCH * STUDENT$$
$$r2 \leftarrow \sigma_{CNO='IT494' \text{ AND } PID='11' \text{ AND } ACADYEAR=2026 \text{ AND } SEM='AUTUMN'}(r1)$$
$$result \leftarrow \pi_{SID,SNAME}(r2)$$

5. Compute course-wise count of Students in Winter 2026 for all elective courses. Produce cols: CNO, CName, NumberOf Students.

$$r1 \leftarrow \sigma_{ACADYEAR=2025 \text{ AND } SEM='WINTER'}(SEMESTER)$$
$$r2 \leftarrow \sigma_{CTYPE='ELECTIVE'}(r1 * OFFERS)$$
$$r3 \leftarrow \sigma_{CTYPE='ELECTIVE'}(r2 * REGISTERS)$$
$$r4(OfferID, StudentCount) \leftarrow {}_{OfferID}\mathcal{F}_{count(SID|*)}(r3)$$
$$result \leftarrow \pi_{CNO,CName,StudentCount}(r4 * OFFERS * COURSE)$$

OR

$$r1 \leftarrow \sigma_{ACADYEAR=2025\ AND\ SEM='WINTER'}(SEMESTER)$$
$$r2 \leftarrow \sigma_{CTYPE='ELECTIVE'}(r1 * OFFERS)$$
$$r3 \leftarrow \sigma_{CTYPE='ELECTIVE'}(r2 * REGISTERS)$$
$$r4(CNO, StudentCount) \leftarrow {}_{CNO}\mathcal{F}_{count(SID|*)}(r3)$$
$$result \leftarrow \pi_{CNO,CName,StudentCount}(r4 * ~~OFFERS * COURSE~~)$$

6. Compute Faculty-wise count of courses in Academic Year 2025-2026. Produce cols: IID, IName, NumberOf Courses.

$$r1 \leftarrow \sigma_{ACADYEAR=2025}(SEMESTER)$$
$$r2 \leftarrow \sigma_{CTYPE='ELECTIVE'}(r1 * OFFERS)$$
$$r3(IID, CourseCount) \leftarrow {}_{IID}\mathcal{F}_{count(OfferID|*)}(r2)$$
$$result \leftarrow \pi_{IID,IName,CourseCount}(r3 * INSTRUCTOR)$$

7. List Students (Cols: SID, SName) having CPI > 7.0 and SPI was never below 6.0

$$r1 \leftarrow \sigma_{SPI<6.0}(RESULT)$$
$$r2 \leftarrow \sigma_{CPI>7.0}\left(STUDENT\ {}_{s}\frac{SEMI-DIFFERENCE}{s.SID=r1.SID}r1\right)$$
$$result \leftarrow \pi_{SID,SName}(r2)$$

OR

$$r1 \leftarrow \pi_{SID}\left(\sigma_{CPI>7.0}(STUDENT)\right)$$
$$r2 \leftarrow \pi_{SID}\left(\sigma_{SPI<6.0}(RESULT)\right)$$
$$r3 \leftarrow (r1 - r2)$$
$$result \leftarrow \pi_{SID,SNAME}(r3 * STUDENT)$$